

Archimedes: A Free Monte Carlo Simulator for Semiconductor Device Simulations

JAI SACHITH PAUL

Archimedes is a free technology computer aided design (TCAD) software package used for designing and simulating submicron and mesoscopic semiconductor devices. Simulation and prediction of the behaviour of devices are based on the ensemble Monte Carlo method. The simulator is extensively used by students for understanding solid-state electronic behaviour, as well as by researchers who want a tested and working transport code to start from.

It is a complete and free software package written in C and can be copied, modified and redistributed under general-public licence (GPL). The GNU project (recursive acronym for GNU's Not Unix) has announced substitution of the software package Aeneas by Archimedes, making this one of the GNU packages for Monte Carlo semiconductor devices simulation. The latest release of the software is version 2.0.1.

Why Archimedes

Researchers working in the semiconductor electronics industry are always interested in the miniaturisation of components. Reduction in size of the components helps in accommodating more components in the same wafer, which increases the computing power per unit area. Simulations are introduced into the design to make accurate guesses of the electron behaviour, thereby reducing costs of prototyping new semiconductor devices.

A very unfortunate situation that exists today is that most of these

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When Archimedes was released, it was a very exciting time. No one in the world at that time would even think about sharing a working code in the semiconductor community. The response from the leader of the GNU project, Richard Stallman, was very positive and that encouraged me to release even more advanced versions. At the moment, Archimedes is utilised by a big community of users and it is well-known in the field. I hope this example shows, to the scientific community in general, that it is important to share scientific and industrial codes. This is our only way to make progress advances. Without sharing the code, we put our society in the situation of duplicating efforts again and again.”

—Jean Michel Sellier, developer of Archimedes



codes for simulation are not freely available. Every university has its own Monte Carlo simulator, a non-equilibrium green's function (NEGF) simulation for similar research. Of course, a very large number of research papers are available, but their codes are not shared. This leads to duplication of work around the world. A researcher has to start from scratch for developing new and more advanced simulations.

Archimedes aims to change this situation in the field of Monte Carlo device simulation by releasing under GPL, allowing the user to download and use it as a starting point of any new program. It is developed as a powerful tool for the simulation of

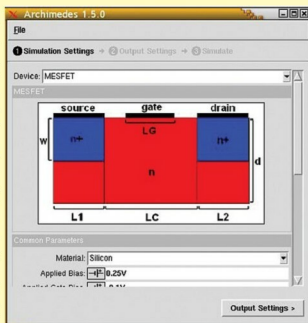


Fig. 1: Archimedes GUI

quite general semiconductor devices. “Many scientific codes, especially in the field of numerical simulations, could greatly benefit the community if released under open licences. This is one way to get serious advancements in the field by eliminating